

Claw GUI Interface

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Chapter 1

README

ClawGUI is a GUI interface to a Raspberry Pico running the claw firmware. ClawGUI is a MacOS application written in Objective-C. This should compile on a stock Xcode install.

Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

AppDelegate()	10
<UIApplicationDelegate>	
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NSObject	
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clawControl	10
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Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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clawControl	10
UART	
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Chapter 4

File Index

4.1 File List

Here is a list of all documented files with brief descriptions:

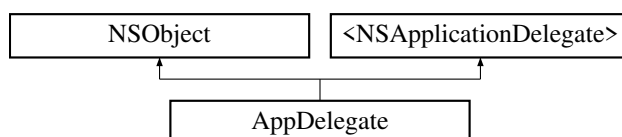
ClawGUI/ AppDelegate.h	17
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Chapter 5

Class Documentation

5.1 AppDelegate Class Reference

Inheritance diagram for AppDelegate:



Instance Methods

- (IBAction) - **connectButtonPressed:**
- (IBAction) - **enableButtonPressed:**
- (IBAction) - **PositionSliderMoved:**
- (IBAction) - **stopClawButtonPressed:**
- (IBAction) - **readStatusButtonPressed:**
- (IBAction) - **speedSliderMoved:**
- (IBAction) - **enableCalibrationButtonPressed:**
- (IBAction) - **setOriginButtonPressed:**
- (IBAction) - **BumpOriginDownButtonPressed:**

Protected Attributes

- dispatch_semaphore_t **serialSem**

The documentation for this class was generated from the following files:

- ClawGUI/AppDelegate.h
- ClawGUI/AppDelegate.m

5.2 AppDelegate() Category Reference

Protected Attributes

- bool **serialPortConnected**
- bool **calibrationModeEnabled**

Properties

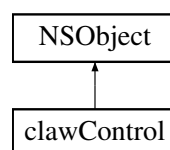
- [clawControl](#) * **claw**
- IBOutlet NSWindow * **window**
- IBOutlet NSComboBox * **serialPortComboBox**
- IBOutlet NSButton * **connectButton**
- IBOutlet NSButton * **enableClawButton**
- IBOutlet NSSlider * **clawPositionSlider**
- IBOutlet NSButton * **stopClawButton**
- IBOutlet NSButton * **readStatusButton**
- IBOutlet NSSlider * **clawSpeedSlider**
- IBOutlet NSButton * **enableCalibrationButton**
- IBOutlet NSButton * **setOriginButton**
- IBOutlet NSButton * **bumpOriginDownButton**

The documentation for this category was generated from the following file:

- ClawGUI/AppDelegate.m

5.3 clawControl Class Reference

Inheritance diagram for clawControl:



Instance Methods

- (NSError *) - [setClawErrorWithCode:andDescription:andReason:andSuggestion:](#)
Creates and NSError with appropriate keys for Claw objects.
- (BOOL) - [setDevicePathWith:error:](#)
Sets device descriptor path to serial port.
- (BOOL) - [connectToSerialPortWithError:](#)
Connects the claw device to the serial port specified with setDevicePathWith:
- (BOOL) - [disconnectFromSerialPortWithError:](#)
Disconnects the claw device from the serial port if already connected.
- (BOOL) - [enableClawStepperWithError:](#)
Sets claw device to enabled.

- (BOOL) - [disableClawStepperWithError:](#)
sets claw device to disabled
- (BOOL) - [setClawPosition:withError:](#)
sets claw position to parameter clawPosition (0 to 100)
- (BOOL) - [stopClawMotionWithError:](#)
stops claw motion immediately, regardless of current position
- (NSDictionary *) - [readStatusToDictionaryWithError:](#)
reads status from claw device and returns dictionary of status
- (BOOL) - [setClawSpeed:withError:](#)
Set claw speed expressed as stepper period.
- (BOOL) - [setClawZeroWithError:](#)
sets claw zero position to current position. Used to calibrate zero on claw
- (BOOL) - [bumpClawZeroDownWithError:](#)
bumps zero and moves stepper 1/4 turn below current zero to bump down zero set point

Protected Attributes

- dispatch_semaphore_t **serialPortSemaphore**
Semaphore for serial port access – NOT IMPLEMENTED, Need to implement for threaded operation.

Properties

- BOOL **connectionStatus**
The status of the serial connection to the Claw controller.
- BOOL **clawStepperEnabled**
The state of the stepper in the claw, Enabled = TRUE, Disabled = False.
- NSString * **devicePath**
The path to the device descriptor in the /dev tree for the serial connection to the Claw controller.

5.3.1 Method Documentation

5.3.1.1 bumpClawZeroDownWithError:

```
- (BOOL) bumpClawZeroDownWithError:
    (NSError**) error
```

bumps zero and moves stepper 1/4 turn below current zero to bump down zero set point

Parameters

<i>error</i>	pointer to NSError* on failure to send bump down command, nil otherwise
--------------	---

Returns

EXIT_FAILURE on fail, EXIT_SUCCESS on success

5.3.1.2 connectToSerialPortWithError:

```
- (BOOL) connectToSerialPortWithError:
    (NSError**) error
```

Parameters

<i>error</i>	pointer to NSError* on failure to connect to the Device, nil otherwise
--------------	--

Returns

EXIT_FAILURE on fail, EXIT_SUCCESS on success

5.3.1.3 disableClawStepperWithError:

```
- (BOOL) disableClawStepperWithError:
    (NSError**) error
```

sets claw device to disabled

Sets claw stepper motor to disabled. Disabling stepper will allow it to spin freely

Parameters

<i>error</i>	pointer to NSError* on failure to disable the claw, nil otherwise
--------------	---

Returns

EXIT_FAILURE on fail, EXIT_SUCCESS on success

5.3.1.4 disconnectFromSerialPortWithError:

```
- (BOOL) disconnectFromSerialPortWithError:
    (NSError**) error
```

Disconnects the claw device from the serial port if already connected.

Disconnects from the claw serial port and frees the file handle

Parameters

<i>error</i>	pointer to NSError* on failure to disconnect to the serial port of the claw device
--------------	--

Returns

returns NSError on failure to disconnect and free file handle

5.3.1.5 enableClawStepperWithError:

```
- (BOOL) enableClawStepperWithError:
    (NSError**) error
```

Sets claw device to enabled.

Sets claw stepper motor to enabled and ready to move claw device

Parameters

<i>error</i>	pointer to NSError* on failure to enable the claw, nil otherwise
--------------	--

Returns

EXIT_FAILURE on fail, EXIT_SUCCESS on success

5.3.1.6 readStatusToDictionaryWithError:

```
- (NSDictionary *) readStatusToDictionaryWithError:
    (NSError**) error
```

reads status from claw device and returns dictionary of status

Status of claw device key Current Position : current claw position in number of steps key Target Position : target position of claw device in number of steps key Step Period : microseconds per step pulse of claw stepper motor key Moving : TRUE if claw is currently moving, FALSE otherwise key Enabled : TRUE if claw is currently enabled, FALSE otherwise key Estop : TRUE if claw E-Stop is currently pressed, FALSE otherwise

Parameters

<i>error</i>	pointer to NSError* on failure to get status
--------------	--

Returns

NSDictionary pointer to dictionary of current status

5.3.1.7 setClawErrorWithCode:andDescription:andReason:andSuggestion:

```
- (NSError *) setClawErrorWithCode:
    (NSInteger) errorCode
    andDescription:(NSString*) errorDescription
    andReason:(NSString*) errorReason
    andSuggestion:(NSString*) errorSuggestion
```

Creates and NSError with appropriate keys for Claw objects.

Parameters

<i>errorCode</i>	NSInteger error code for error
<i>errorDescription</i>	NSString pointer to description of error
<i>errorReason</i>	NSString pointer to the reason the error occurred
<i>errorSuggestion</i>	NSString pointer to a suggestion for the user to rectify the error

Returns

: returns NSError created or nil if failed to create error

5.3.1.8 setClawPosition:withError:

```
- (BOOL) setClawPosition:
    (NSInteger) clawPosition
    withError:(NSError**) error
```

sets claw position to parameter clawPosition (0 to 100)

Sets claw position based on clawPosition parameter to a value between 0 and 100

Parameters

<i>clawPosition</i>	NSInteger position
<i>error</i>	pointer to NSError* on failure set the claw position, nil otherwise

Returns

EXIT_FAILURE on fail, EXIT_SUCCESS on success

5.3.1.9 setClawSpeed:withError:

```
- (BOOL) setClawSpeed:
    (NSInteger) clawPeriod
    withError:(NSError**) error
```

Set claw speed expressed as stepper period.

Note

Claw speed is expressed as a period in microseconds

Sets claw speed in microseconds per step. Reasonable speeds are between 40µs and 400µs (25kHz to 2.5kHz)

Parameters

<i>clawPeriod</i>	microsecond period to set claw stepper motor pulse rate
<i>error</i>	pointer to NSError* on failure to get status

Returns

NSDictionary pointer to dictionary of current status

5.3.1.10 setClawZeroWithError:

```
- (BOOL) setClawZeroWithError:
    (NSError**) error
```

sets claw zero position to current position. Used to calibrate zero on claw

Parameters

<i>error</i>	pointer to NSError* on failure to send set zero command, nill otherwise
--------------	---

Returns

EXIT_FAILURE on fail, EXIT_SUCCESS on success

5.3.1.11 setDevicePathWith:error:

```
- (BOOL) setDevicePathWith:
    (NSString*) pathToSet
    withError:(NSError**) error
```

Sets device descriptor path to serial port.

Parameters

<i>pathToSet</i>	NSString pointer to device path of serial port
<i>error</i>	pointer to NSError* on failure to set device path, nil otherwise

Returns

NSError if pathToSet parameter is nil

5.3.1.12 stopClawMotionWithError:

```
- (BOOL) stopClawMotionWithError:
    (NSError**) error
```

stopps claw motion immediately, regardless of current position

Parameters

<i>error</i>	pointer to NSError* on failure to send stop command, nill otherwise
--------------	---

Returns

EXIT_FAILURE on fail, EXIT_SUCCESS on success

The documentation for this class was generated from the following files:

- ClawGUI/clawControl.h
- ClawGUI/clawControl.m

5.4 UART Struct Reference

Structure to hold [UART](#) data and state, include termios options.

Public Attributes

- int **uartFileHandle**
File handle for [UART](#) device descriptor.
- char * **uartPath**
Absolute path to [UART](#) file descriptor, should be in /dev/.
- int **uartBaud**
[UART](#) Baud Rate.
- int **uartParity**
[UART](#) Parity (Even/Odd/None).
- int **uartStopBits**
[UART](#) Stop Bits.
- int **uartByteSize**
[UART](#) Byte Size, should be 8.
- struct termios **uartOptions**
termios [UART](#) Options, This is where the actual options get stored.

5.4.1 Detailed Description

Structure to hold [UART](#) data and state, include termios options.

The documentation for this struct was generated from the following file:

- ClawGUI/clawControl.m

File Documentation

```
00001 //
00002 // AppDelegate.h
00003 // ClawGUI
00004 //
00005 // Created by Jon Wade on 12/23/25.
00006 //
00007
00008 #import <Cocoa/Cocoa.h>
00009
00010 @interface AppDelegate : NSObject <NSApplicationDelegate>
00011
00012 {
00013     dispatch_semaphore_t serialSem;
00014 }
00015
00016
00017 - (IBAction)connectButtonPressed:(id) sender;
00018
00019 - (IBAction)enableButtonPressed:(id) sender;
00020
00021 - (IBAction)PositionSliderMoved:(id) sender;
00022
00023 - (IBAction)stopClawButtonPressed:(id) sender;
00024
00025 - (IBAction)readStatusButtonPressed:(id) sender;
00026
00027 - (IBAction)speedSliderMoved:(id) sender;
00028
00029 - (IBAction)enableCalibrationButtonPressed:(id) sender;
00030
00031 - (IBAction)setOriginButtonPressed:(id) sender;
00032
00033 - (IBAction)BumpOriginDownButtonPressed:(id) sender;
00034
00035 @end
00036
```

```

00001
00012 #import <Foundation/Foundation.h>
00013
00014
00015 #define CLAW_MAX_PERIOD                400                // Maximum claw
    step period Freq = 1 / period = 2.5kHz
00016 #define CLAW_MIN_PERIOD                40                // Minimum claw
    step period Freq = 1 / period = 25kHz
00017
00018 #define CLAW_ERROR_DOMAIN              @"com.jon-wade.ClawGUI.SerialError"    // NSError domain
    for errors created in clawControl.c
00019 #define CLAW_ERROR_INVALID_PORT_PATH  100                // Error code for
    invalid serial port path

```

```

00020 #define CLAW_ERROR_NO_DATA_TO_READ          101          // Error code for
        no data in receive queue of serial port
00021 #define CLAW_ERROR_STEPPER_NOT_CONNECTED    200          // Error code for
        claw controller not connected to serial port
00022 #define CLAW_ERROR_STEPPER_NOT_ENABLED      201          // Error code for
        claw controller stepper motor not connected
00023 #define CLAW_ERROR_INCORRECT_RESPONSE      202          // Error code for
        incorrect response to command, specifically status command
00024
00025 NS_ASSUME_NONNULL_BEGIN
00026
00027 @interface clawControl : NSObject
00028 {
00029     dispatch_semaphore_t serialPortSemaphore;
00030 }
00031
00032 @property BOOL connectionStatus;
00033
00034 @property BOOL clawStepperEnabled;
00035
00036 @property (strong) NSString* devicePath;
00037
00038 - (NSError*) setClawErrorWithCode: (NSInteger) errorCode
00039               andDescription: (NSString*) errorDescription
00040               andReason: (NSString*) errorReason
00041               andSuggestion: (NSString*) errorSuggestion;
00042
00043 - (BOOL) setDevicePathWith: (NSString*) pathToSet withError: (NSError**) error;
00044
00045 - (BOOL) connectToSerialPortWithError: (NSError**) error;
00046
00047 - (BOOL) disconnectFromSerialPortWithError: (NSError**) error;
00048
00049 - (BOOL) enableClawStepperWithError: (NSError**) error;
00050
00051 - (BOOL) disableClawStepperWithError: (NSError**) error;
00052
00053 - (BOOL) setClawPosition: (NSInteger) clawPosition withError: (NSError**) error;
00054
00055 - (BOOL) stopClawMotionWithError: (NSError**) error;
00056
00057 - (NSDictionary*) readStatusToDictionaryWithError: (NSError**) error;
00058
00059 - (BOOL) setClawSpeed: (NSInteger) clawPeriod withError: (NSError**) error;
00060
00061 - (BOOL) setClawZeroWithError: (NSError**) error;
00062
00063 - (BOOL) bumpClawZeroDownWithError: (NSError**) error;
00064
00065 @end
00066
00067 NS_ASSUME_NONNULL_END

```

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